

Part 2 – Secure VPC Network and Firewall Security

Google Cloud Security Engineering Lab

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Domain: Cloud Security Engineering

Platform: Google Cloud Platform (GCP)

1. Objective

The objective of this project is to design and implement a secure Virtual Private Cloud (VPC) environment in Google Cloud Platform.

The project demonstrates:

- Custom VPC creation
- Multi-region subnet configuration
- Firewall rule implementation
- Virtual machine deployment
- Secure network communication
- Connectivity validation

2. Architecture Overview

A custom VPC network was created with two regional subnets.

Component	Details
VPC Network	vpc-network-zgs3
Subnet A	10.10.10.0/24
Subnet B	10.10.20.0/24
VM 1	us-test-01
VM 2	us-test-02

Architecture Diagram

Secure VPC Network Architecture – Part 2 (UML Diagram)

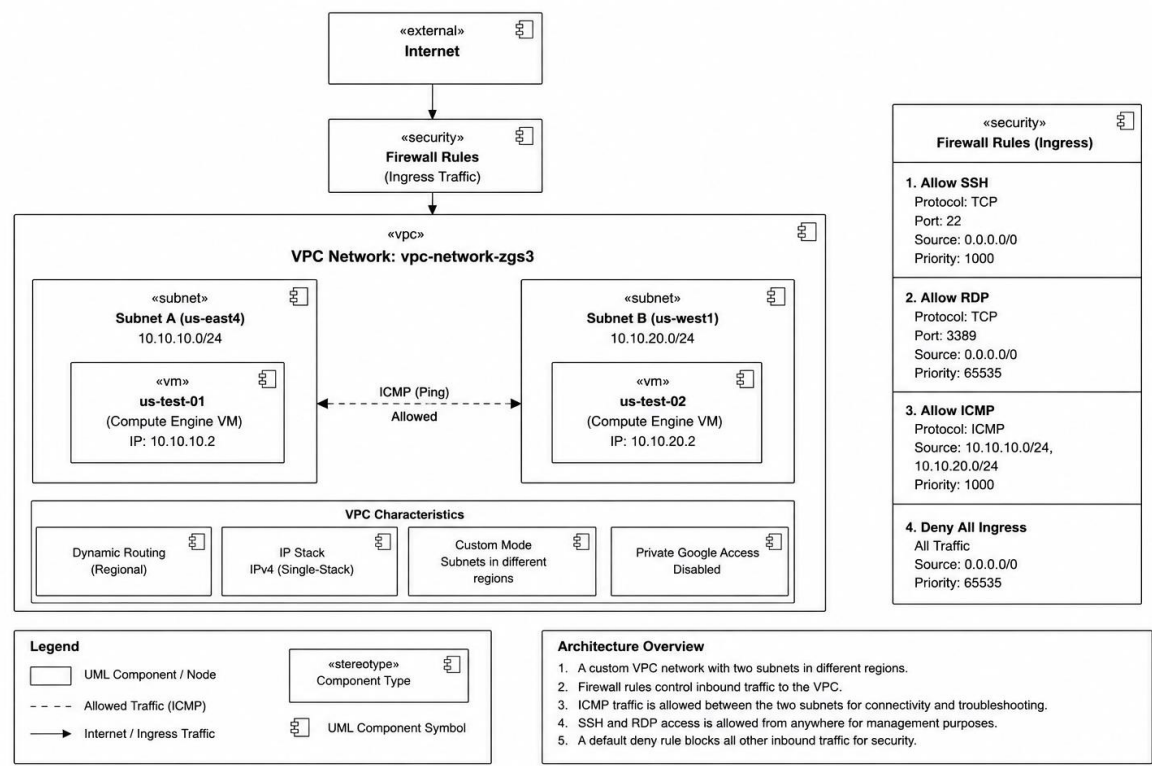


Figure 1: Overall VPC Network Architecture

3. VPC Network Creation

A custom VPC network was created to provide network isolation and secure communication between cloud resources.

Screenshot

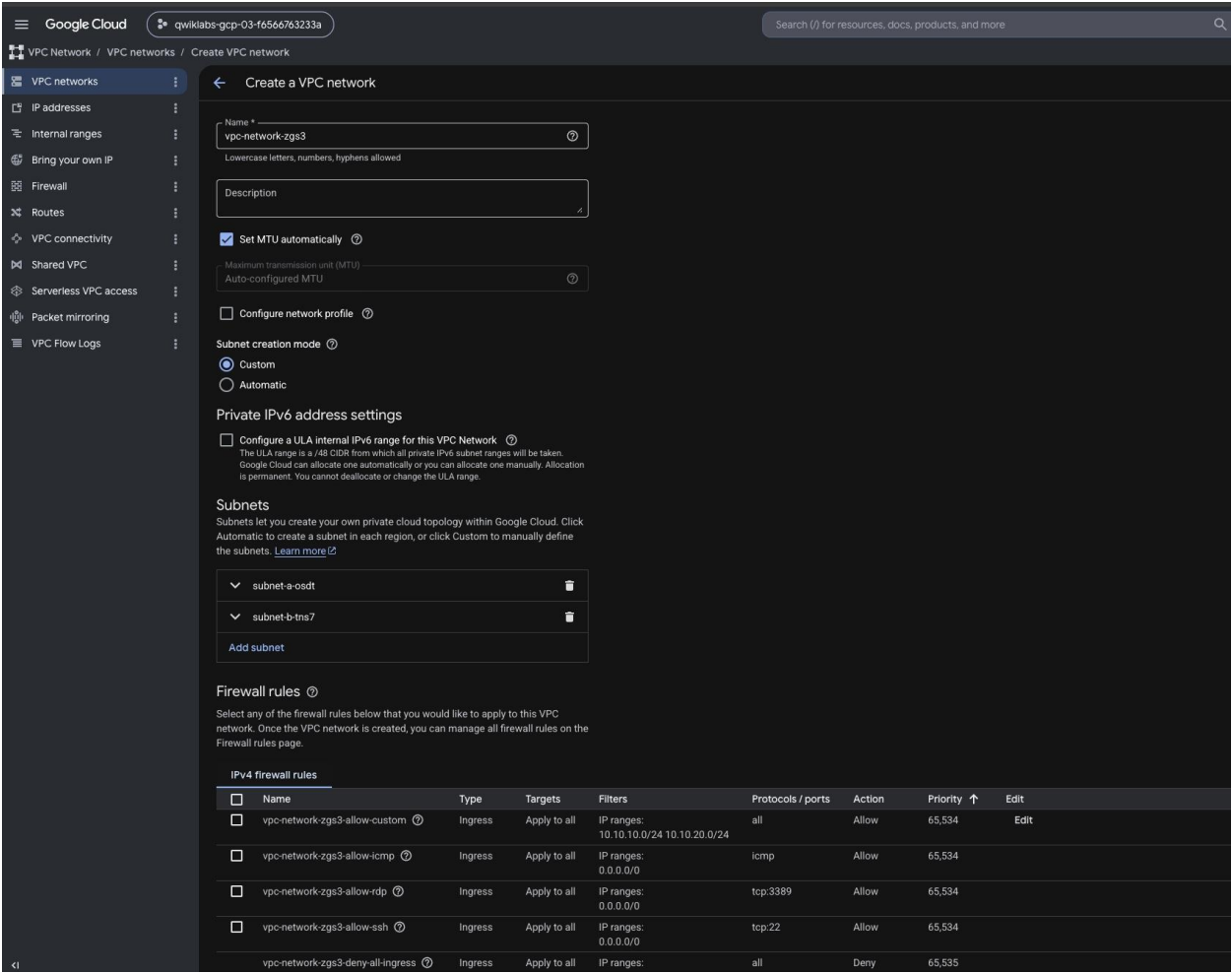


Figure 2: Custom VPC Network Creation

Description

The VPC was configured in Custom Mode, allowing manual subnet management and greater control over network segmentation.

4. Subnet Configuration

Two subnets were created in different Google Cloud regions.

Subnet Details

Subnet	Region	CIDR Range
subnet-a-osdt	us-east4	10.10.10.0/24
subnet-b-tns7	us-west1	10.10.20.0/24

Google Cloud

quickstart-gcp-03-f6566b3223a

Search (2) for resources, docs, products, and more

VPC network / VPC networks / Network: vpc-network-zgs3

VPC networks

IP addresses

Internal ranges

Bring your own IP

Firewall

Routes

VPC connectivity

Shared VPC

Serverless VPC access

Packet mirroring

VPC Flow Logs

VPC network details

Delete VPC network

vpc-network-zgs3

Overview

Subnets

Static internal IP addresses

Firewalls

Endpoint groups

Firewall endpoints

Routes

VPC network peering

Private services access

DNS configuration

Subnets

Add subnet

Manage flow logs

Filter

Enter property name or value

Name	Region	Stack Type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal ranges	FQDN	Gateway	Private Google Access	Flow log configs
subnet-a-zgs3	us-east4	IPv4 (single-stack)	10.10.10.0/24			None	None	10.10.10.1	Off	
subnet-b-zgs3	us-west1	IPv4 (single-stack)	10.10.20.0/24			None	None	10.10.20.1	Off	

Reserved proxy-only subnets for load balancing

Name	Region	IP address ranges	Gateway	Role	Purpose
No rows to display					

Figure 3: Subnet Configuration

Description

The subnets were deployed across separate regions to demonstrate regional network segmentation and routing.

5. Firewall Configuration

Firewall rules were configured to control inbound network traffic.

Configured Rules

Rule	Protocol	Port
SSH	TCP	22
RDP	TCP	3389
ICMP	ICMP	N/A
Deny-All	All	All

Screenshot

Deployment groups
Endpoint groups
Secure Web Proxy
Web Proxies
SWP Policies
URL Lists
DNS Armor
Threat detection
Common components
Authentication Configuration
Address groups
Security profiles
TLS inspection policies
SSL policies

Refresh
Configure logs
Delete

Filter
Enter property name or value

☐	Name	Type	Targets	Filters	Protocols / ports	Action	Priority	Network	Logs	Hit count	Last hit	Insights
☐	default-allow-icmp	Ingress	Apply to all	IP ranges: 0.0.0.0/0	icmp	Allow	65534	default	Off	--	--	
☐	default-allow-internal	Ingress	Apply to all	IP ranges: 10.128.0.0/9	tcp:0-65535 udp:0-65535 icmp	Allow	65534	default	Off	--	--	
☐	default-allow-dns	Ingress	Apply to all	IP ranges: 0.0.0.0/0	tcp:3389	Allow	65534	default	Off	--	--	
☐	default-allow-ssh	Ingress	Apply to all	IP ranges: 0.0.0.0/0	tcp:22	Allow	65534	default	Off	--	--	
☐	ovsx-firewall-ssh	Egress	Apply to all	IP ranges: 0.0.0.0/0	tcp:22	Allow	1000	ovsx-network-zg3	Off	--	--	
☐	ovsx-firewall-udp	Egress	Apply to all	IP ranges: 0.0.0.0/24	tcp:3389	Allow	65535	ovsx-network-zg3	Off	--	--	
☐	ovsx-firewall-icmp	Ingress	Apply to all	IP ranges: 10.10.20.0/24, 10.10.10.0/24	icmp	Allow	1000	ovsx-network-zg3	Off	--	--	

Network firewall policies

Firewall policies let you group several firewall rules so that you can update them all at once, effectively controlled by Identity and Access Management (IAM) roles.

[Learn more](#)

Refresh

Figure 4: Firewall Rule Configuration

Description

The firewall rules allow administrative access while maintaining network security through controlled ingress traffic.

6. Virtual Machine Deployment

Two Compute Engine virtual machines were deployed within separate subnets.

Screenshot

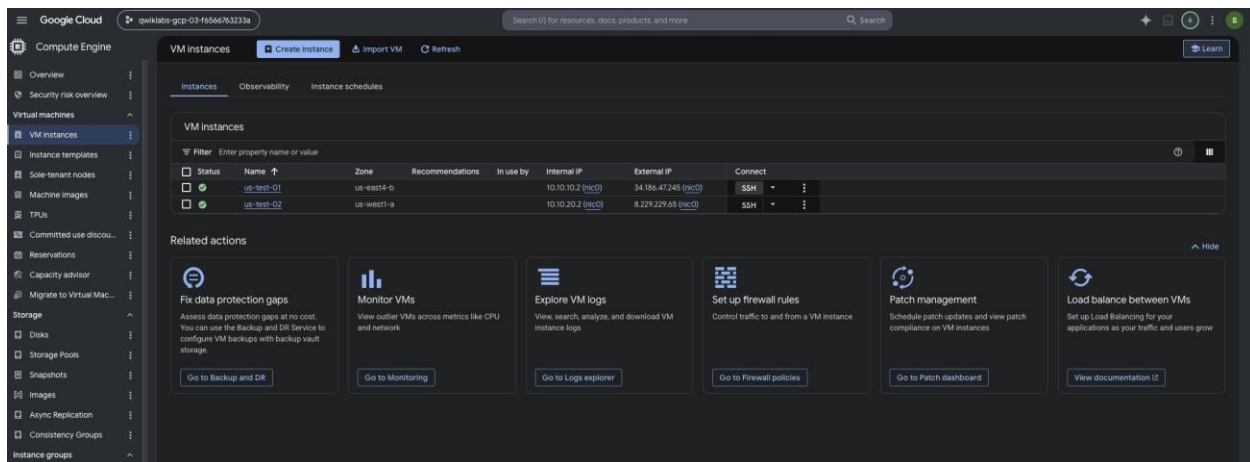


Figure 5: Virtual Machine Deployment

Description

The VMs were used to validate network communication and firewall configurations.

7. Connectivity Testing

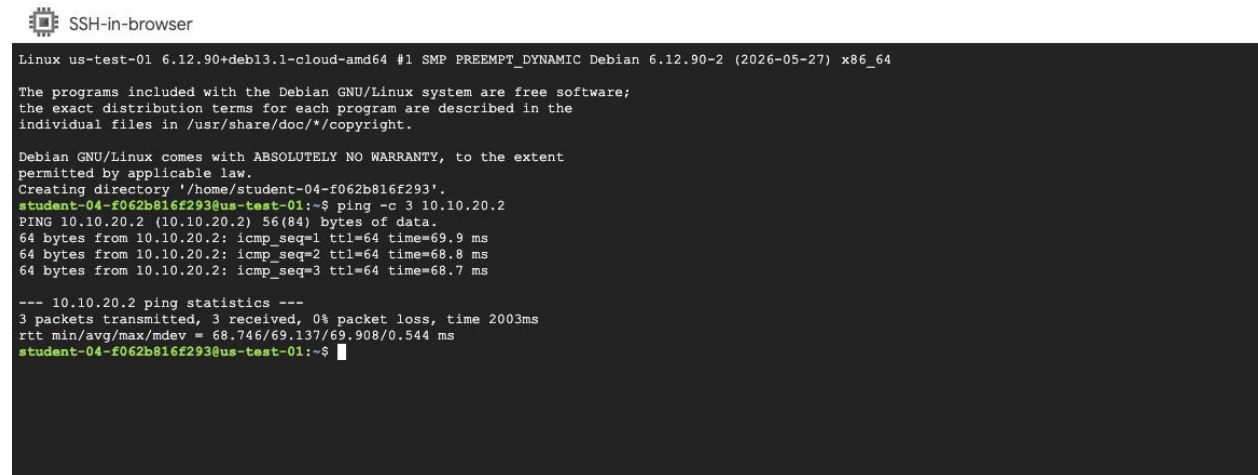
Connectivity testing was performed between the two virtual machines.

Ping Test

Command:

```
ping -c 3 10.10.20.2
```

Screenshot



```
SSH-in-browser

Linux us-test-01 6.12.90+deb13.1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.90-2 (2026-05-27) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory '/home/student-04-f062b816f293'.
student-04-f062b816f293@us-test-01:~$ ping -c 3 10.10.20.2
PING 10.10.20.2 (10.10.20.2) 56(84) bytes of data.
64 bytes from 10.10.20.2: icmp_seq=1 ttl=64 time=69.9 ms
64 bytes from 10.10.20.2: icmp_seq=2 ttl=64 time=68.8 ms
64 bytes from 10.10.20.2: icmp_seq=3 ttl=64 time=68.7 ms

--- 10.10.20.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 68.746/69.137/69.908/0.544 ms
student-04-f062b816f293@us-test-01:~$
```

Figure 6: Successful ICMP Connectivity Test

Result

- 3 packets transmitted
- 3 packets received
- 0% packet loss

Analysis

The successful ICMP response confirms routing and firewall configuration between both subnets.

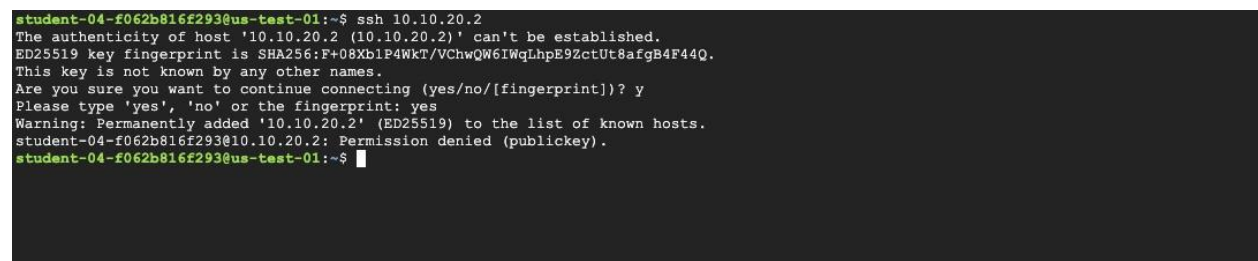
8. SSH Validation

SSH connectivity was tested between the virtual machines.

Command:

```
ssh 10.10.20.2
```

Screenshot



```
student-04-f062b816f293@us-test-01:~$ ssh 10.10.20.2
The authenticity of host '10.10.20.2 (10.10.20.2)' can't be established.
ED25519 key fingerprint is SHA256:F+08Xb1P4WkT/VChwQW6IWqLhpE9ZctUt8afgB4F44Q.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '10.10.20.2' (ED25519) to the list of known hosts.
student-04-f062b816f293@10.10.20.2: Permission denied (publickey).
student-04-f062b816f293@us-test-01:~$
```

Figure 7: SSH Connectivity Validation

Result

Permission denied (publickey)

Analysis

The connection successfully reached the SSH service. Authentication failed because no valid key was provided. This confirms that the firewall rule allowed TCP port 22 traffic.

9. Security Concepts Demonstrated

The following cloud security concepts were implemented:

- Network Segmentation
- Firewall Security
- Secure Remote Administration
- Access Control
- Infrastructure Isolation
- Connectivity Validation
- Defense in Depth

10. Lessons Learned

- Learned how VPC networks are designed in Google Cloud.
- Gained practical experience creating regional subnets.
- Implemented and tested firewall rules.
- Understood how cloud routing works between subnets.
- Validated network security through connectivity testing.
- Improved troubleshooting and cloud administration skills.

11. Conclusion

This project successfully demonstrated the design and implementation of a secure Google Cloud VPC environment.

The implementation included custom networking, subnet segmentation, firewall security controls, virtual machine deployment, and network validation testing. The project provided

hands-on experience with core cloud security and networking concepts commonly used in enterprise environments.